

Alan Bailey

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EDUCATION

Rice University

B.S. in Electrical and Computer Engineering

- **GPA:** 4.0/4.0

Houston, TX

August 2025 – Present

Westlake Boys High School

Class rank 1 for final 3 years

Auckland, New Zealand

February 2020 – December 2024

RELEVANT COURSEWORK

Calculus I, Calculus II, Multivariable Calculus, Differential Equations, Honors Mechanics, Honors Electromagnetism, Computer Engineering, Computational Thinking

EXPERIENCE

Translational Bioelectronics Lab (Seymour Lab)

May 2026 – July 2026

Undergraduate Researcher - **Automated Wafer Defect Detection**

Houston, TX

- Trained a YOLOv11 defect detection model from scratch, achieving **mAP50** $\approx$ **0.75** on semiconductor wafer imagery, then integrated it with a Gemini Vision LLM to build a full automated inspection pipeline
- Designed a confidence-based routing system that pre-filters images with YOLO before calling the LLM, cutting API inference by **90%** and reducing end-to-end inspection time by **5x**
- Built a Streamlit review dashboard used by the lab for ongoing wafer inspections, allowing researchers to validate YOLO bounding boxes and correct model labels to build a retraining dataset

Rice EXP LAB

January 2026 – May 2026

Undergraduate Research Engineer

Houston, TX

- Design and testing a high-fidelity wind system.
- Prototyping an array of 3D-printed servo-actuated wind ducts.
- Creating a system to translate 3D engine data into physical airflow through microcontroller-controlled PWM fans.

Rice Eclipse

August 2025 – Present

Avionics Flight Engineer

Houston, TX

- Select and design the control electronics and sensors for an Active Flight Stabilisation system for rockets to increase altitude.
- Designed and launched a Level 1 certification rocket which reached 3700ft. Modeled and 3D printed a nosecone which housed an altimeter.

Rice Wind Energy

August 2025 – Present

Blades Sub Team Engineer

Houston, TX

- Research airfoils and simulate various blade configurations using Qblades to maximise turbine power production.
- Evaluate the strength of different composites and manufacturing methods to optimise the turbine blade's structure, while balancing material sustainability.

ADDITIONAL COURSES

IBM Qiskit Global Summer School | Qiskit IBM

July 2025

- Developed and simulated quantum circuits using Qiskit, completing all 4 labs
- Explored quantum error mitigation and correction techniques, circuit optimization, and algorithm implementation.

CS50 Introduction to Artificial Intelligence with Python | HarvardX (edX)

February 2025 – March 2025

- Completed 12 Python projects applying AI concepts such as search algorithms, reinforcement learning, and neural networks.

TECHNICAL SKILLS

Languages: Python

3D design: Fusion360, Bambu Studio

Libraries: pandas, NumPy, Matplotlib, Qiskit, Pillow

Machine Learning: YOLOv11 / Ultralytics, VLMs, Computer Vision, Human-in-the-loop ML